§6.4 The Gram-Schmidt Process

Example 2 in textbook.

Take $x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$.

$W = \text{span}\{x_1, x_2, x_3\}$ is a subspace of $\mathbb{R}^4$.

Step 1 $v_1 = x_1$

Step 2 $v_2 = x_2 - \dfrac{\langle x_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1$ $= \text{proj}_{W_1} x_2$, $W_1 = \text{span}\{v_1\}$.

$$= \cdots = \frac{1}{4} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Step 3 $v_3 = x_3 - \dfrac{\langle x_3, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 - \dfrac{\langle x_3, v_2 \rangle}{\langle v_2, v_2 \rangle} v_2$

$= -\text{proj}_{W_2} x$, $W_2 = \text{span}\{v_1, v_2\}$.

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \frac{\left\langle \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{\left\langle \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \frac{1}{4} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle}{\left\langle \frac{1}{4} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \frac{1}{4} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle} \frac{1}{4} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \frac{2}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{\cancel{\frac{1}{4}}(2)}{\frac{1}{\cancel{4}\,2}(12)} \left(\frac{1}{\cancel{4}} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{6} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \cdots =$$

Check final answer.

Theorem 11 The Gram-Schmidt Process

Suppose $\{x_1, \dots, x_k\}$ is a basis for a subspace W of $\mathbb{R}^n$ (or $\mathbb{C}^n$). Define

$$v_1 = x_1$$

$$v_2 = x_2 - \frac{\langle x_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1$$

$$\vdots$$

$$v_j = x_j - \left(\sum_{\ell=1}^{j-1} \frac{\langle x_j, v_\ell \rangle}{\langle v_\ell, v_\ell \rangle} v_\ell \right)$$

$\text{proj}_{W_j} x_j$

$W_j = \text{span}\{v_1, \dots, v_{j-1}\}$

Then $\{v_1, \dots, v_k\}$ is an orthogonal basis of W.

Moreover, $\text{span}\{x_1, \dots, x_j\} = \text{span}\{v_1, \dots, v_j\}$ for all $1 \le j \le k$.

In particular $\text{span}\{v_1, \dots, v_k\} = \text{span}\{x_1, \dots, x_k\} = W$.

Remarks

- Above $\{v_1, \dots, v_k\}$ is an orthogonal basis for W. If want an <u>orthonormal</u> basis for W, then set $w_j = \frac{1}{\|v_j\|} v_j$. Then $\{w_1, \dots, w_k\}$ is an orthonormal basis for W.

- Once we have an orthonormal basis $B = \{w_1, \dots, w_k\}$ for W

then any $y \in W$ has the form

$$y = \langle y, w_1 \rangle w_1 + \cdots + \langle y, w_k \rangle w_k \quad (\text{Thm } 5, \S 6.2)$$

and $$[y]_B = \begin{bmatrix} \langle y, w_1 \rangle \\ \vdots \\ \langle y, w_k \rangle \end{bmatrix} \in \mathbb{R}^k \ (\text{or } \mathbb{C}^k)$$

Moreover we can compute $\operatorname{proj}_W x$ for any $x \in \mathbb{R}^n$ using the formula $\operatorname{proj}_W x = \sum_{j=1}^{k} \langle x, w_j \rangle w_j$.

Example

Take $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} i \\ 1 \end{bmatrix} \right\}$ to be a basis for $\mathbb{C}^2$.

Let's check $$\left\langle \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} i \\ 1 \end{bmatrix} \right\rangle = 1(\overline{i}) + (0)(\overline{1})$$
$$= -i \neq 0.$$

Using Gram-Schmidt.

Set $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Set $$v_2 = \begin{bmatrix} i \\ 1 \end{bmatrix} - \frac{\left\langle \begin{bmatrix} i \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\rangle} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} i \\ 1 \end{bmatrix} - \frac{(i)(\overline{1}) + (1)(\overline{0})}{1^2} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} i \\ 1 \end{bmatrix} - i \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Now $\{v_1, v_2\} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} i \\ 1 \end{bmatrix} \right\}$ is an orthogonal basis.

If we had mistakenly taken the inner product $\left\langle \begin{bmatrix} i \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\rangle$ backwards, then $\left\langle \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} i \\ 1 \end{bmatrix} \right\rangle = (1)\overline{(i)} + (0)\overline{(1)}$

$$= -i$$

and we would compute $\begin{bmatrix} i \\ 1 \end{bmatrix} - (-i)\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} i \\ 1 \end{bmatrix} + \begin{bmatrix} i \\ 0 \end{bmatrix}$

$$= \begin{bmatrix} 2i \\ 1 \end{bmatrix}.$$

$\left\langle \begin{bmatrix} 2i \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\rangle = 2i \neq 0$. Bad !! .

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§ 6.7 Inner product spaces

Until now we have looked at the standard inner products on $\mathbb{R}^n$ and $\mathbb{C}^n$.
What about other spaces?

Def. Let V be a vector space over $\mathbb{R}$ (or $\mathbb{C}$).
An <u>inner product</u> on V is a function $\langle \cdot, \cdot \rangle$ which takes pairs of vectors in V to scalars in $\mathbb{R}$ (or $\mathbb{C}$). In addition the function satisfies

1. $\langle u, v \rangle = \overline{\langle v, u \rangle} \quad \forall\, u, v \in V.$

2. $\langle u+v, w\rangle = \langle u,w\rangle + \langle v,w\rangle \quad \forall u,v,w \in V.$

3. $\langle cu, v\rangle = c\langle u,v\rangle \quad \forall c \in \mathbb{R}$ (or $\mathbb{C}$), $u,v \in V$.

4. $\langle u,u\rangle \geq 0$ and $\langle u,u\rangle = 0 \iff u = 0$ (zero vector)

A vector space with an inner product is called an <u>inner product space</u>.

Examples $\mathbb{R}^n$ and $\mathbb{C}^n$ with the standard inner product are inner products.

Theorem

Suppose V is an inner product space.
Then $\langle w, u+v\rangle = \langle w,u\rangle + \langle w,v\rangle$
and $\langle w, cv\rangle = \bar{c}\langle w,v\rangle \quad \forall c \in \mathbb{R}$ or $\mathbb{C}$.

Proof of second equation.

$$\langle w, cv\rangle = \overline{\langle cv, w\rangle} = \overline{c\langle v,w\rangle}$$

$$= \bar{c}\,\overline{\langle v,w\rangle}$$

$$= \bar{c}\langle w,v\rangle$$

Example 1 in textbook.

Define $\langle \cdot,\cdot\rangle' : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$.

by $\left\langle \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \right\rangle' = 4u_1v_1 + 5u_2v_2$.

Let check $\langle u, v\rangle' = \langle v, u\rangle'$ (Property 1 in def'n of inner product)

$$\left\langle \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \right\rangle' = 4u_1v_1 + 5u_2v_2$$

$$= 4v_1u_1 + 5v_2u_2$$

$$= \left\langle \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}, \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \right\rangle'.$$

To check property 4 in the def'n of inner product.

$$\left\langle \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \right\rangle' = 4u_1^2 + 5u_2^2 \geq 0$$ since $4u_1^2$ and $5u_2^2$ are ≥ 0.

This proves that the first half of Property 4 in the def'n is satisfied by $\langle \cdot, \cdot \rangle'$. For the second part:

$$0 = \left\langle \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \right\rangle' \iff 0 = u_1^2 + u_2^2.$$

$$\iff u_1^2 = 0 \text{ and } u_2^2 = 0 \quad (\text{since } u_1^2, u_2^2 \geq 0)$$

$$\iff u_1 = u_2 = 0$$

$$\iff \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The other properties are also satisfied. This is a long and boring exercise.

Exercise. Show that replacing 4 and 5 in the def'n of $\langle \cdot, \cdot \rangle'$ with $a, b > 0$ give us an inner product on $\mathbb{R}^2$.

Example 2 in textbook

Take $V = \mathbb{P}_n$ = polynomials of degree n or less with real coefficients

Pick $n+1$ distinct real numbers $t_0, \dots, t_n \in \mathbb{R}$.

Define $\langle p, q \rangle = p(t_0)q(t_0) + \cdots + p(t_n)q(t_n)$

for any $p, q \in \mathbb{P}_n$.

Let's check Property 3 in the def'n of inner product.

Let $c \in \mathbb{R}$ and $p, q \in \mathbb{P}_n$.

Then
$$\begin{aligned}\langle cp, q \rangle &= (cp(t_0))q(t_0) + \cdots + (cp(t_n))q(t_n) \\ &= c(p(t_0)q(t_0)) + \cdots + c(p(t_n)q(t_n)) \\ &= c\left(p(t_0)q(t_0) + \cdots + p(t_n)q(t_n)\right) \\ &= c\langle p, q \rangle.\end{aligned}$$

The summands are all products of real numbers.

Let's check $\langle p, p \rangle \geq 0$ (Property 4 in def'n)

$$\langle p, p \rangle = (p(t_0))^2 + \cdots + (p(t_n))^2$$
$$\geq 0 \quad \text{since } p(t_j)^2 \geq 0 \ \forall\, 0 \leq j \leq n.$$

Finally $0 = \langle p, p \rangle \iff 0 = (p(t_0))^2 + \cdots + (p(t_n))^2$

$\iff p(t_0) = \cdots = p(t_n) = 0.$

$\Rightarrow$ p has $n+1$ distinct roots (zeros)

$\Rightarrow$ $p = 0$ since $\deg p \leq n$.